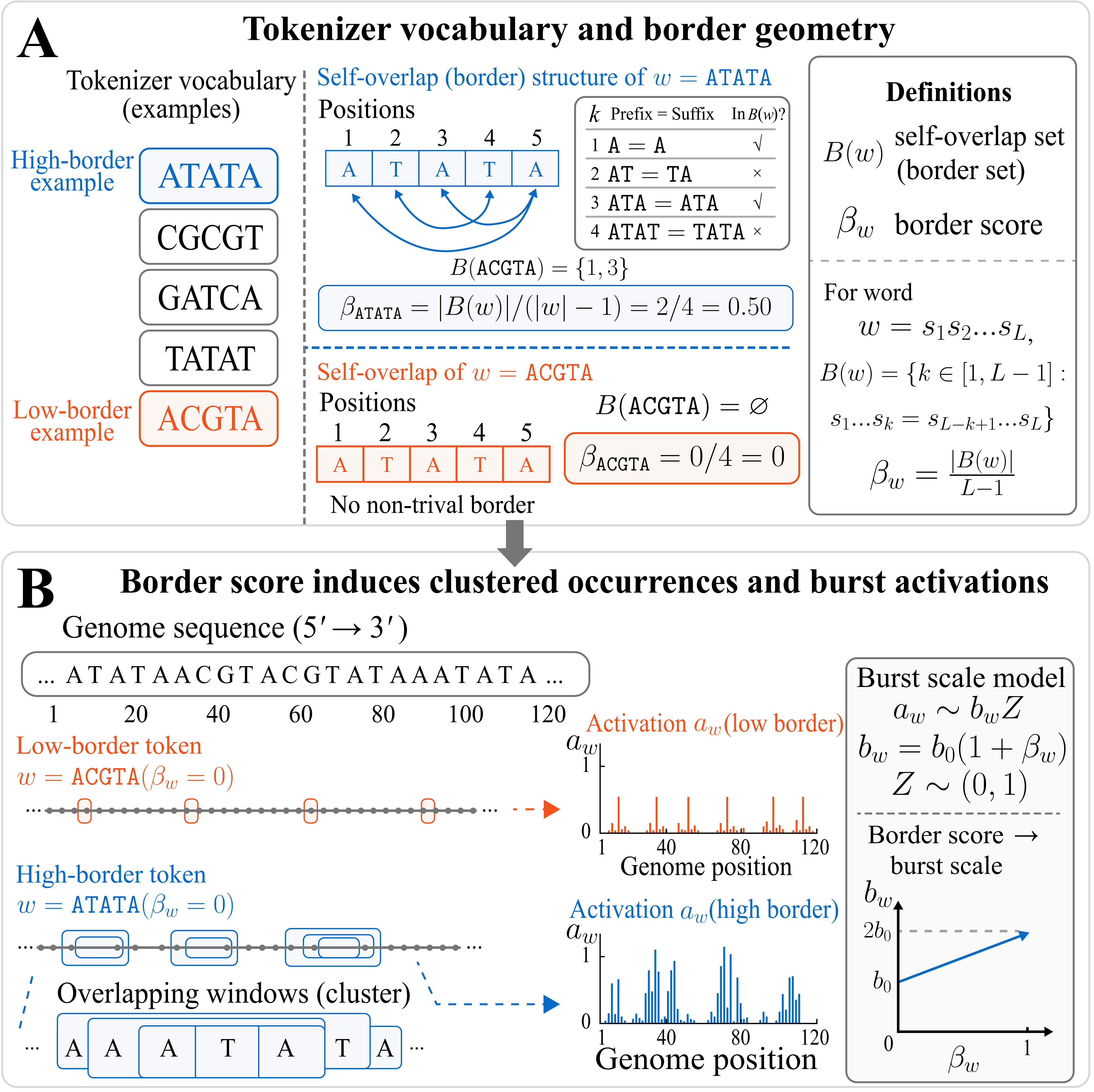

A
Tokenizer vocabulary and border geometry
Tokenizer vocabulary (examples)
High-border example
ATATA
CGCGT
GATCA
TATAT
Low-border example
ACGTA
Self-overlap (border) structure of w = ATATA
Positions
1 2 3 4 5
A T A T A
k Prefix = Suffix In B(w)?
1 A = A √
2 AT = TA ×
3 ATA = ATA √
4 ATAT = TATA ×
B(ACGTA) = {1,3}
β_ATATA = |B(w)|/(|w| − 1) = 2/4 = 0.50
Self-overlap of w = ACGTA
Positions
1 2 3 4 5
A T A T A
B(ACGTA) = ∅
β_ACGTA = 0/4 = 0
No non-trival border
Definitions
B(w) self-overlap set (border set)
β_w border score
For word
w = s1s2...sL,
B(w) = {k ∈ [1, L − 1] : s1...sk = sL−k+1...sL}
β_w = |B(w)| / (L−1)
B
Border score induces clustered occurrences and burst activations
Genome sequence (5′ → 3′)
... A T A T A A C G T A C G T A T A A A T A T A ...
1 20 40 60 80 100 120
Low-border token
w = ACGTA(β_w = 0)
High-border token
w = ATATA(β_w = 0)
Overlapping windows (cluster)
... A A A T A T A ...
Activation a_w(low border)
Activation a_w(high border)
a_w
1
0
1 40 80 120
Genome position
Burst scale model
a_w ∼ b_w Z
b_w = b_0(1 + β_w)
Z ∼ (0, 1)
Border score → burst scale
b_w
2b_0
b_0
0
1
β_w